



Liquid fuels demand and supply model

Analyst kickoff

Coverage: SACU + priority African regions

2 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Durban | vopak.com



Vopak provides infrastructure that connects supply to markets

ABOUT VOPAK | Storage and handling infrastructure serving energy and manufacturing



What matters about the client

Business model: storage and handling across oil, gas, chemicals and industrial terminals. [1]

Customers: producers, manufacturers, distributors, governments and traders. [1]

South Africa: coastal Durban and inland Lesedi connect to the New Multi-Product Pipeline (NMPP). [3]

Strategy: predictable cash flows, stronger assets and disciplined growth investment. [2]

Implication for this engagement

Translate fuel flows into terminal throughput, handling and usable storage needs.

Establish whose volumes could use each corridor, access needs and competing options.

Assess coastal receipts, inland flows and terminal needs within the national fuel system.

Test sustained demand, utilisation and commercial access before calling a gap an opportunity.

This assignment: South Africa outlook | Namibia / Walvis Bay assessment | Priority African market screening. [4]

[1] Vopak: Services

[2] Vopak: Who we are / strategy

[3] Vopak: South Africa expansion

Vopak's strategy provides three lenses for African fuel infrastructure

Published group priorities | application to our study is our interpretation

	What Vopak reports	What this means for the African study
Improve	Improve financial and sustainability performance of existing assets. Target annual operating cash returns of 13–17%.	South Africa: assess how changing national fuel flows affect terminal utilisation, handling demand, connectivity and operating performance.
Grow	Prioritise gas and industrial terminals with attractive, long-term contracted returns. Group investment ambition: EUR 2.6bn through 2030.	Walvis Bay / Namibia: test customer commitments, competing capacity and corridor access. Fuel deficits alone do not establish a viable terminal investment.
Accelerate	Selective energy-transition investment: EUR 1.4bn ambition by 2030, including low-carbon fuels, sustainable feedstocks, ammonia and liquid CO ₂ .	Priority African markets: screen sustained service demand and transition compatibility. The agreed model scope remains petrol, diesel and jet.

The report provides group strategy; it does not set out a dedicated Africa expansion plan or a Walvis Bay investment commitment.

Commercial lens: sustained customer demand, usable infrastructure, contracted returns and investment requirements.

African fuel opportunities depend on sustained demand for storage and handling

Fuel flows matter when they support sustained demand for terminal capacity and services

Storage services

EUR 551.0m | 81.4%

H1 2026 consolidated revenue

Product movements

EUR 40.4m | 6.0%

H1 2026 consolidated revenue

Storage / handling related

EUR 68.9m | 10.2%

H1 2026 consolidated revenue

Other services

EUR 16.8m | 2.5%

H1 2026 consolidated revenue

All four categories reconcile to EUR 677.1m; displayed shares total 100.1% because of rounding.

Study implication: connect usable tank capacity and holding time to receipts, dispatch and throughput.

South Africa

Assess changing coastal receipts and inland flows across the national system.

Client output: service demand, utilisation pressures and constraints for existing and potential terminals.

Walvis Bay / Namibia

Test the customer volumes and corridors that could sustain a storage and handling proposition.

Client output: accessible demand, competing facilities, usable capacity and commercial conditions.

Priority African markets

Screen for sustained fuel flows, customer needs and commercially accessible infrastructure.

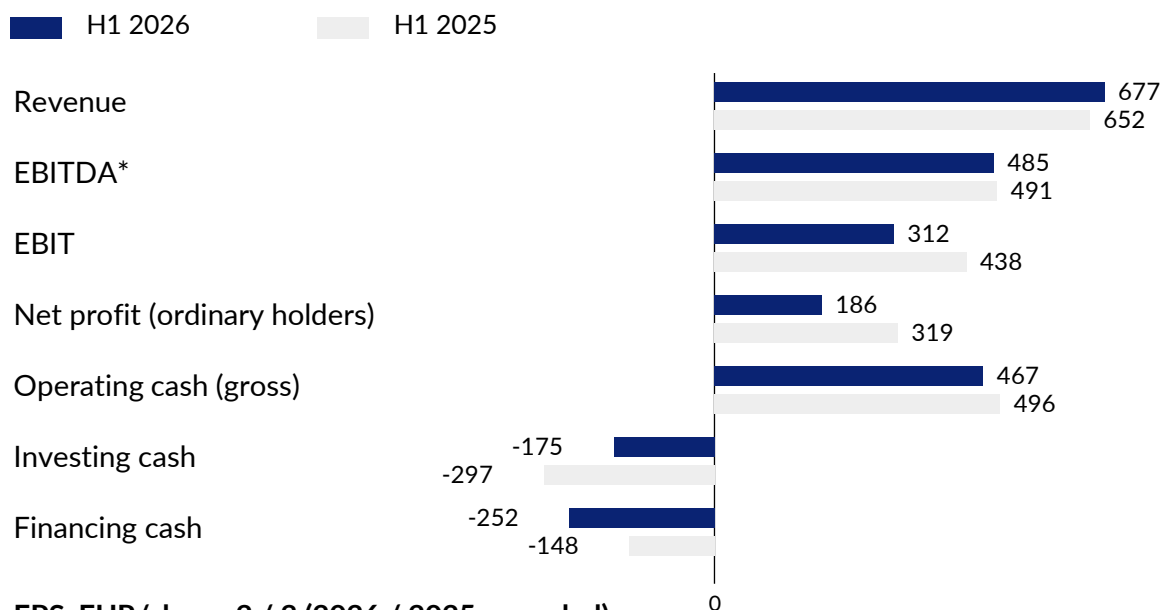
Client output: markets to prioritise for detailed asset and customer assessment.

Commercial test: sustained customer volumes, compatible capacity, handling capability, access and contract terms.

Group revenue growth alone does not establish attractive African terminal economics

Vopak H1 2026 | six months to 30 June | EUR million unless stated | rounded to whole numbers

Group income and cash flows



*EBITDA excludes exceptional items (APM). EBIT and net profit include them. H1 2025 profit included the AVTL listing gain.

Study implication: assess contracted capacity, handling demand and costs in South Africa; test customer commitments at Walvis Bay; screen commercially accessible African flows.

Revenue by service

Storage services

551 | 81%

Product movements

40 | 6%

Storage / handling related

69 | 10%

Other services

17 | 2%

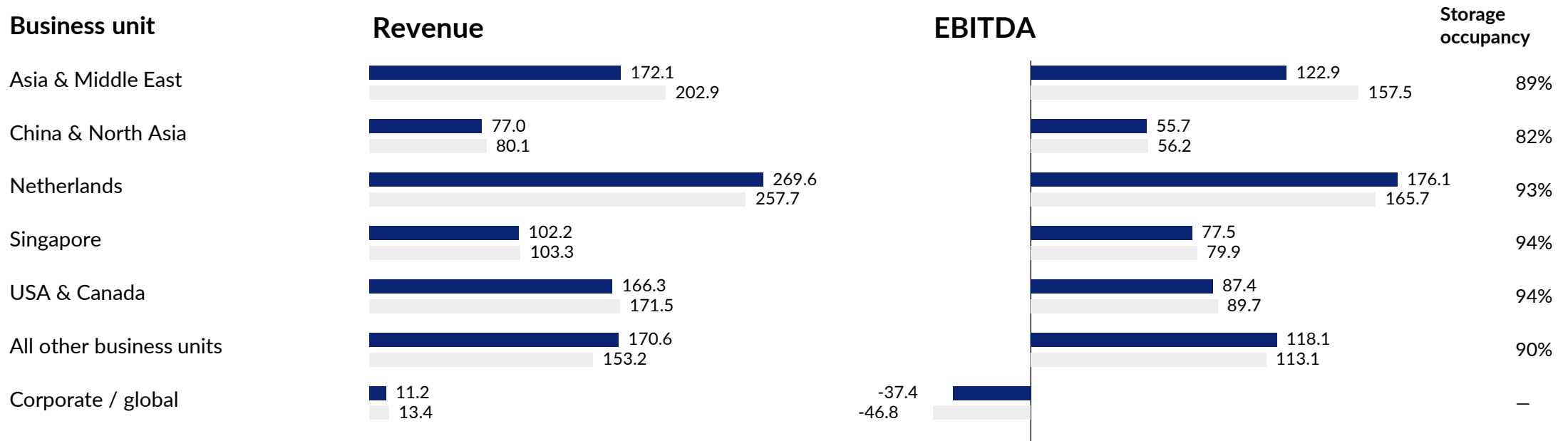
Service mix is revenue, not profit. Storage and handling earnings are not disclosed separately.

Regional earnings provide context; African opportunities need local asset economics

Vopak H1 2026 | six months to 30 June | EUR million unless stated

■ H1 2026 ■ H1 2025

Proportional results, excluding exceptional items | common scale within each panel



Group: revenue 969.0 / 982.1; EBITDA 600.3 / 615.3 (2026 / 2025); 2026 occupancy 91%.

Study implication: assess South African terminal utilisation and access; test Walvis Bay customer economics; compare priority African markets using local evidence, not group margins.

Storage occupancy = average capacity rented / average storage capacity, weighted by ownership. Other business units is not a separate African result.

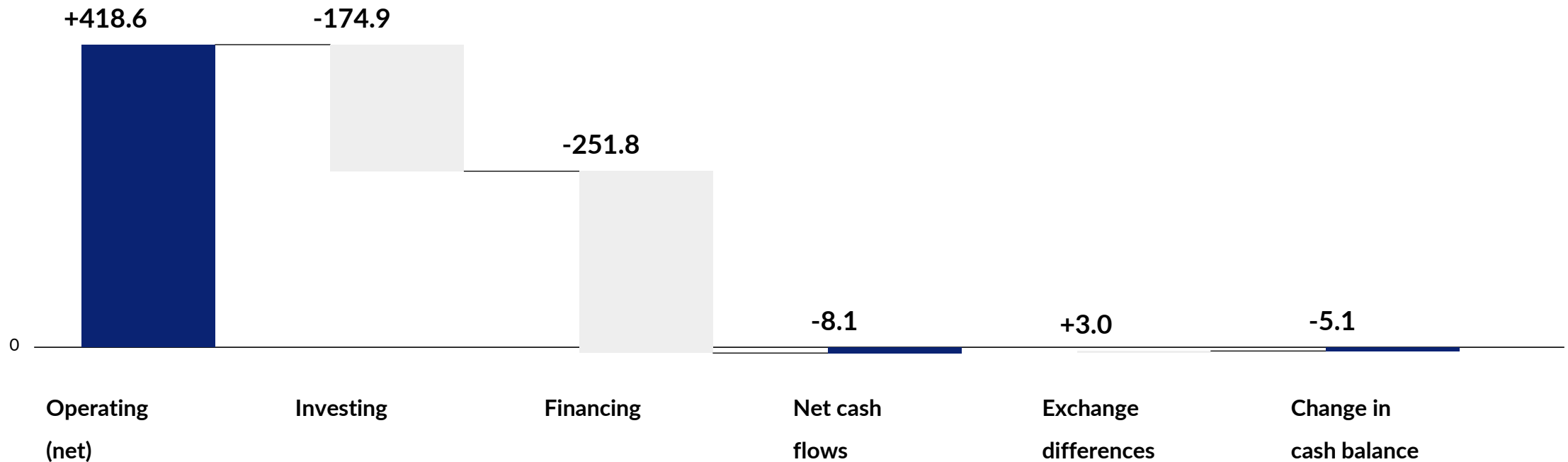
Source: Royal Vopak, Half Year Report 2026; pp. 23–25, 56–57. Interpretation is our analytical framing.

Vopak generates operating cash; investment and financing absorb that cash

Vopak H1 2026 | six months to 30 June | EUR million unless stated

Net operating cash flow = 466.6 gross + 5.4 interest received – 53.4 tax paid = 418.6

Net operating: EUR 418.6m / 454.9m. Proportional operating free cash flow: EUR 444.2m / 450.5m (2026 / 2025; separate APM).



Study implication: test service cash generation in South Africa, investment and customer commitments at Walvis Bay, and funding requirements in priority African markets.

Group operating cash is positive; total cash flow is –8.1m. Exchange +3.0m gives a –5.1m change: 98.8m opening → 93.7m closing.

Source: Royal Vopak, Half Year Report 2026; pp. 32, 57. Consolidated cash balances include bank overdrafts; proportional free cash flow is a separate APM.

Align scope and delivery expectations

A shared starting point

Align the team on the project question, required outputs and six-week priorities.

A model and infrastructure build brief

Connect the fuel model with transport, terminal and storage capacity.

An evidence-led working discussion

Show inherited data, Python results, discrepancies and the checks still needed.

A basis for delivery decisions

Agree the first baseline, workstream interfaces, review gates and responsibilities.

What this document is not

A validated market forecast

Current results include provisional assumptions and unresolved differences.

An investment recommendation

An import gap alone does not establish an accessible storage opportunity.

A completed infrastructure model

Maps show evidence and connectivity; usable capacity and access still need validation.

A commitment to equal depth everywhere

Regional screening and detailed country modelling have different evidence requirements.

Connect changing fuel flows to Vopak's asset and growth decisions

Where do changing liquid-fuel flows across South Africa, Walvis Bay and priority African markets create opportunities for Vopak?

01 Project question	Where can changing South African liquid-fuel balances support Vopak storage and handling services; does Walvis Bay offer a credible opportunity; which African markets merit follow-up?
02 Analytical specifications	Map petrol, diesel and jet flows across South African ports, inland markets and infrastructure; assess customer demand, competing terminal operators, usable capacity and commercial access.
03 Model build	Build detailed South African fuel and infrastructure calculations; add focused Namibia/Walvis Bay analysis and SACU/African screening. Apply results to Vopak assets and growth options.
04 Delivery approach	Over six weeks, develop the board narrative: South African market changes, Vopak asset and growth opportunities, Walvis Bay and African priorities.

The test: where can Vopak serve sustained customer demand through its existing assets or credible growth options?

Frame the South African outlook, Walvis Bay case and African priorities



Situation

Vopak needs an updated South African liquid-fuels outlook, a focused assessment of Namibia and Walvis Bay, and a broader view of investment opportunities across Africa.



Complication

Changing domestic production and imports can redirect coastal and inland fuel flows. Route constraints, competing operators and customer access determine which volumes Vopak can serve.



Resolution

Update the South African liquid-fuels outlook, assess Namibia's storage, handling and corridor opportunities, and screen priority African markets for further consideration.

The project requires both market modelling and commercial analysis.

Determine which changing fuel flows Vopak could store and handle

- ② How will changing petrol, diesel and jet demand, domestic production and imports affect liquid-fuel balances and demand for Vopak storage and handling services?

01 Market outlook

How much must be imported after domestic production, and where could available imports still leave demand unmet?

02 Commercial relevance

Which customer flows could Vopak store and handle through accessible terminals and corridors, considering competing operators?

Define the markets, products and scenarios the model must cover

Specification area	Scope and approach
Products and outputs	Petrol, diesel and jet demand, total supply, import requirements and balances
Geographic coverage	SACU; priority markets in East, West, Central and North Africa and SADC excluding SACU*
Time horizon	Use the existing 2024-2050 horizon. Agree near-term decision checkpoints
Scenarios	Independent H/M/L demand and supply; M/M baseline
Evidence and acceptance	Reconcile history; trace inputs; record review and limitations

Demand-supply matrix

Rows: demand / Columns: supply

Demand / Supply	H	M	L
H	H/H	H/M	H/L Tight supply
M	M/H	M/M Baseline	M/L
L	L/H Ample supply	L/M	L/L

H = High M = Medium L = Low

Supply: domestic production + available imports

*Regional screening uses high-level data. Detailed country modelling follows agreed priorities and data availability; depth varies by market.

Trace how the major assumptions change petrol, diesel and jet demand

South Africa | selected 2030 drivers + 2024 sector anchors; full inventory separate; M unset

Demand driver	H demand	M demand	L demand	Fuel affected / mechanism
GDP per capita, 2015 ZAR	78,347	Unset	76,061	Petrol, diesel, jet / GDP links
EV share of new vehicles	1.03%	Unset	4.27%	Petrol, diesel / ICE additions
Efficiency gain: petrol / diesel	1.0% / 0.5%	Unset	1.5% / 1.0%	Petrol, diesel / new cohorts
Departing passengers, million	27.06	Unset	27.06	Jet / shared passenger path
OCGT load factor	70.0%	Unset	16.0%	Diesel / power generation
Industry: 2024 anchor	2.50 bn L	Unset	2.50 bn L	Diesel / GDP elasticity 1
Agriculture: 2024 anchor	0.70 bn L	Unset	0.70 bn L	Diesel / GDP elasticity 0.2

Resulting demand, billion litres in 2030 (provisional)

Output	H demand	M demand	L demand	Coverage
Petrol	8.95	Unset	8.53	Petrol 95
Diesel	17.30	Unset	13.17	50 ppm; marine 500 ppm excluded
Jet	2.79	Unset	2.35	Jet A1

Industry/agriculture anchors are provisional shared inputs, not 2030 forecasts. Annual km: 17k / 28k / 70k (car / LCV / HCV).

Identify the demand inputs to validate and the evidence still missing

Active vintage: 2026 draft | module-level inventory; full declarations remain in the input register

Module / fuel	Assumptions used	H/L treatment	Evidence / status
Vehicles / petrol, diesel	GDP per capita; sales regression; EV uptake; fleet cohorts; scrappage; mileage; fuel and segment mix; efficiency	GDP, EV and efficiency vary; other parameters shared	Workbook regression / EV inputs dated 24 Mar 2025. Fleet/activity parameters provisional; opening stock unresolved.
Aviation / jet	GDP per capita; departing passengers; regression coefficients; jet allocation; declared monthly profile	GDP varies; passengers and coefficients shared	Workbook RegJetFuel / PaxBaseScenario, 24 Mar 2025. Departures are the current proxy; arrivals, cargo and flight movements are not separately modelled. Validate fuel coverage; monthly profile not wired.
Power / diesel	OCGT MW; load factor; efficiency; energy per litre; operating days; commissioning coverage	Load factor varies; capacity and conversions shared	Workbook named ranges, 24 Mar 2025. Dispatch and commissioning need validation; load-shedding series not wired.
Industry / diesel	Base-year volume; GDP scaling; elasticity; declared product / monthly allocation	Baseline / elasticity shared; macro path varies	Provisional baseline and elasticity; no verified inventory/date; declared monthly profile not wired.
Agriculture / diesel	Base-year volume; GDP scaling; elasticity; product allocation	Baseline / elasticity shared; macro path varies	Provisional placeholders; hectares/activity driver still to build.
Marine / diesel, fuel oil	Base-year volume; GDP scaling; elasticity; product split	Baseline / split shared; macro path varies	Provisional baseline and split; port/bunker evidence still needed. Fuel oil shown separately.

Full values: companion Assumption Inventory PDF. All six modules require review; M is unset.

Establish domestic production before assessing import requirements

South Africa | sources reviewed 2 Oct 2026 | utilisation columns are 2030 model assumptions

Asset	Latest sourced status*	Model capacity, kbpd	2030 H util.	2030 M util.	2030 L util.
ENREF	Terminal conversion announced; Oct status unverified [1]	120	0.0%	Unset	0.0%
SAPREF	Redevelopment programme [2]	180	0.0%	Unset	0.0%
NATREF	Disruption reported; Oct availability unverified [3]	108.5	85.0%	Unset	70.0%
SASOL	Secunda operating in FY2026 [4]	75	94.4%	Unset	87.8%
ASTRON	Mar maintenance; Oct availability unverified [5]	100	90.0%	Unset	80.0%
PETROSA	GTL reinstatement planned [6]	45	0.0%	Unset	0.0%

Domestic output, bn litres | 90% availability; asset-specific product yields

Product	H domestic	M domestic	L domestic	Available imports
Petrol	7.42	Unset	6.59	Not modelled
Diesel	2.93	Unset	2.55	Not modelled
Jet	0.95	Unset	0.81	Not modelled

[1] Engen, Apr 2021

[2] CEF, 9 Sep 2026

[3] Sasol, 1 Sep 2026

[4] Sasol, 1 Sep 2026

[5] DMPR, 10 Mar 2026

[6] CEF, 23 Sep 2026

*Published evidence, not confirmation of today's availability. H/L production remains linked to demand; M and imports are unset.

Distinguish required imports from imports that can actually be delivered

South Africa | billion litres | workbook imports and Python import requirements

Fuel	2022 recorded imports	Excel 2030 H	Excel 2030 L	Python 2030 H	Python 2030 L
Petrol	5.46	2.56	4.39	1.53	1.94
Diesel	11.91	11.60	9.66	15.15	11.39
Jet	1.06	2.32	1.72	1.84	1.53

H/L identify demand cases with linked production assumptions; import volumes can move differently by fuel.

Fuel	Python minus Excel: H	Python minus Excel: L	Comparison in both cases
Petrol	-1.03	-2.44	Excel higher
Diesel	+3.55	+1.73	Python higher
Jet	-0.47	-0.19	Excel higher

Differences use unrounded values. Python shows demand less domestic production, including both diesel grades. Exports, stock changes and deliverable import limits remain to add. Neither model yet runs the independent nine-case matrix.

Workbook history is unverified; Notes B7 flags partial 2023 coverage, so 2022 is shown. Excel forecasts contain external workbook links.

Define the infrastructure inputs needed to test feasible fuel flows

Required inputs by asset class | sourced assumptions to build, not validated capacities

Asset class	Required assumptions	Model connection
Ports	Berths, vessel / parcel limits, discharge rates, operating windows and terminal connections	Import receipts and export dispatch
Airports	Jet demand allocation, supply routes, tankage, hydrant / receipt limits and stock cover	Aviation fuel delivery and inventory
Rail	Fuel-compatible links, train paths, wagons, payload, turnaround and service reliability	Feasible rail fuel throughput
Roads	Routes, tanker payloads, fleet availability, trip times, access and operating restrictions	Feasible road fuel throughput
Pipelines	Product / gas type, direction, links, throughput, availability and shared capacity	Liquid-fuel flows; gas dependencies separate
Power lines	Grid connections, transfer limits, outages and supply reliability at relevant assets	Constraints on asset operations and backup demand
Power stations	Fuel / technology, MW, commissioning, availability, dispatch and efficiency	Fuel demand and electricity supply context
Tanks / storage	Gross vs working capacity, segregation, commitments, stock cover and handling limits	Usable inventory and handling constraints

Use source, vintage, units, owner and access basis for every input. Missing capacity remains unknown.

Fuel transport: m3/year | Storage: m3 | Electricity: MW / MWh. Do not convert power-line capacity directly into fuel throughput.

Translate fuel balances into accessible storage and handling opportunities

Purpose: connect the national fuel outlook to evidence of demand for Vopak services

Step	Question to resolve	Output for the next step
01 National balance	How much fuel must be imported after domestic production, exports and stock movements?	Import requirement by fuel and period; compare with available supply to identify the unmet gap
02 Feasible flows	Which ports and inland routes can actually deliver the available volumes?	Feasible flows and bottlenecks after transport, power and terminal-capacity constraints
03 Commercial access	Which customer flows could Vopak serve, given contracts, competitors and access?	Customer volumes accessible to Vopak; assumptions and evidence gaps remain explicit
04 Service needs	What receipt, handling, dispatch and inventory needs follow from those flows?	Terminal throughput, handling needs and usable storage; implications for existing assets and growth

Each step needs evidence: required imports are not guaranteed delivery. Throughput is a flow; storage is a stock.

Organise the model build across five connected workstreams

WS0 Governance: priorities, integration, ownership, review and acceptance

Workstream	Role in the integrated model
WS1 Data validation	Source inputs, reconcile history and record evidence quality and gaps
WS2 Fuel model	Develop demand, production, imports and the scenario framework
WS3 Reporting and delivery	Partner owns the storyline; analyst develops supporting evidence and exhibits
WS4 Analyst enablement	Establish a documented working process for repeatable runs and refreshes
WS5 Infrastructure and logistics	Model ports, airports, rail, roads, pipelines, storage and power dependencies

Integration across WS2 / WS5 / WS3: Nigel leads; Manish builds and reconciles; Henry reviews.

Check shared products, geography, periods and units; conserve volumes; avoid double counting; trace every exhibit to its run.

Reconcile the fuel model and build the infrastructure layer



Situation

We inherit an Excel workbook, prior market analysis and a Python fuel model with repeatable runs and recorded inputs.



Complication

The fuel baseline is not reconciled, scenarios need development, and the national balance does not show which routes and terminals can serve demand.



Resolution

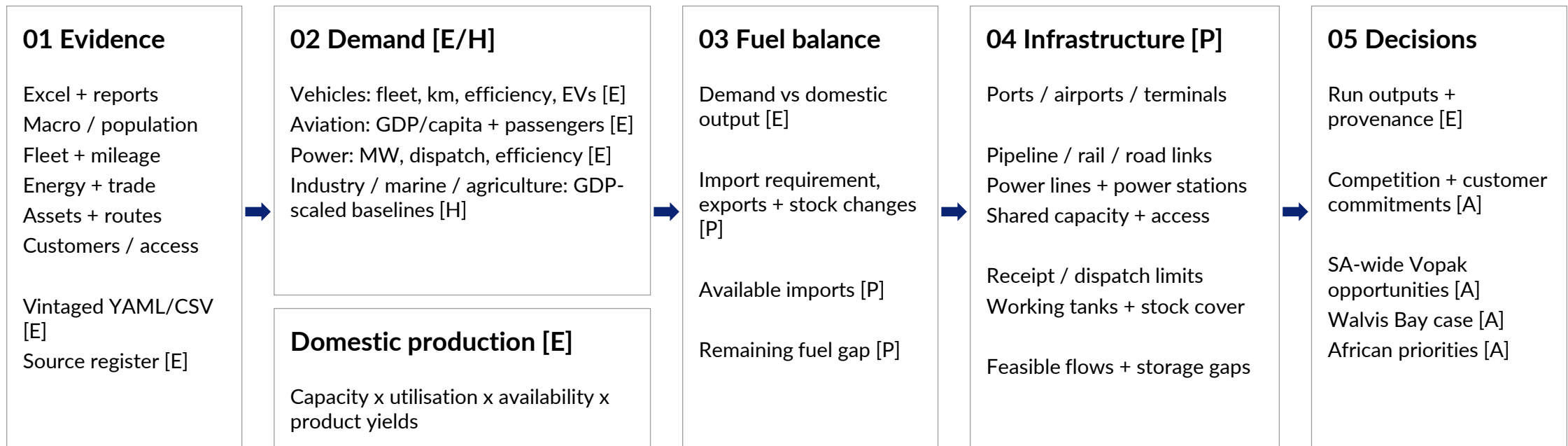
Validate the fuel model and add a connected infrastructure layer. Govern inputs, calculations and outputs so the team can reproduce and review the results.

One integrated design; separate fuel, infrastructure and client-analysis responsibilities.

Connect demand, production, imports and infrastructure in one model design

Whole-model design | [E] existing, provisional [H] held baseline [P] planned [A] engagement analysis

Shared dimensions: market / country x fuel x time x scenario | SA populated; SACU + priority Africa extensions [P]

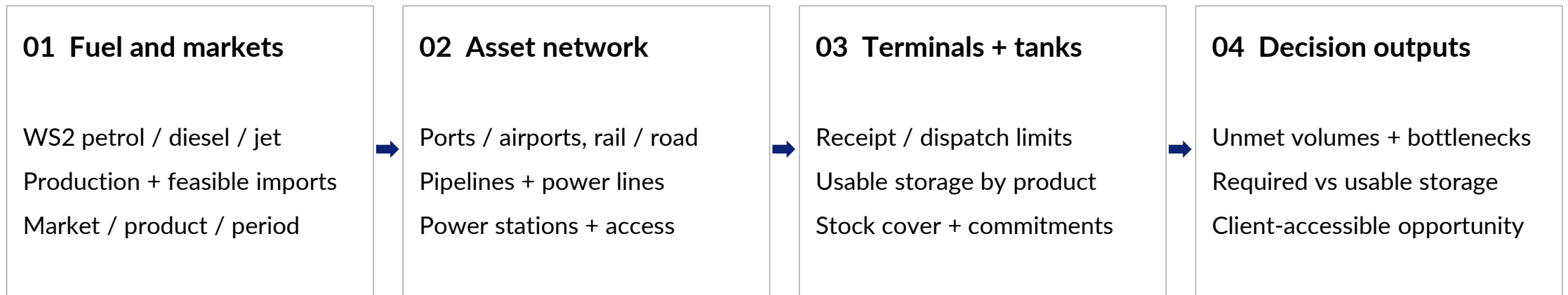


Scenarios across the chain: two paired runs [E] -> independent H/M/L demand x available supply; M/M baseline [P]

Controls across the chain: input resolution, code/input hashes, checks and exceptions [E]; reconciliation, review and approval remain open.

Test which fuel flows the infrastructure can transport, handle and store

WS5 | Liquid-fuel transport demand: allocate petrol, diesel and jet balances to feasible routes



Transport and handling = flow (m3/year) | Storage = stock (m3)

Required working storage = annual throughput / days in year x stock-cover days.

Power lines and power stations inform asset availability and fuel demand; keep MW / MWh separate from fuel volumes.

First release: South Africa inventory + one evidenced Durban-to-Gauteng route. Unknown capacity remains unassessed.

Resolve the baseline before extending scenarios and market coverage

Build priority	Evidence needed to move forward
1 Establish the baseline	Explain Excel/Python differences; record the reference run and source the first infrastructure inventory
2 Complete fuel scenarios	Validate demand and production drivers; define imports and independent H/M/L cases with M/M baseline
3 Prove one infrastructure route	Reconcile route, handling and storage limits; demonstrate a feasible flow with no shared-capacity double counting
4 Extend the evidence	Agree market tiers: detailed South Africa; focused Namibia/Walvis Bay; SACU and wider African screening. Record evidence gaps and escalation criteria
5 Integrate decision outputs	Link balances to terminal implications and accessible opportunities; reproduce results and record review findings

These dependencies shape the six-week delivery plan that follows; sourcing and storyboarding start immediately.

Coordinate model development, infrastructure analysis and reporting over six weeks

Proposed six-week programme | fuel model, infrastructure and client storyline progress together

Activity / workstream	W1	W2	W3	W4	W5	W6
WS1/2 Baseline and Excel reconciliation						
WS2 Drivers, imports and H/M/L scenarios						
WS5 Asset inventory and usable capacity						
WS5 First route, handling and tank limits						
WS2/5 Integrated flows and sensitivities						
WS3 Walvis Bay, competitors, market screening						
WS3 Partner-led storyline and evidence						
WS0/3 Review and board pack / handover						

W1 gate: reproducible baseline + sourced infrastructure inventory + first corridor agreed

W3: first route calculation | W4: integrated draft | W5: review findings | W6: board pack

Apply common review gates to the fuel and infrastructure models

Common acceptance criteria | Manish prepares evidence; Nigel reviews methodology; Henry challenges readiness

Review gate	Demand and supply	Infrastructure
1 Establish the baseline	Source inputs; reconcile Excel–Python differences; document production and imports	Source asset inventory; identify capacity, connectivity and access gaps
2 Demonstrate calculations	Trace demand drivers and production calculations; check balances	Demonstrate one corridor; check flows, handling limits and storage requirements
3 Test scenarios and outputs	Check scenario responses, import requirements and reproducible outputs	Check feasible flows, bottlenecks and storage needs under those scenarios

Record checks, unresolved gaps and review findings at every gate. A passing run is not validation or approval.

Timing sits on the Gantt. These are development review gates; formal-use requirements remain in GATE_CHECKLIST.md.

Source: WS5 brief and integrated six-week workplan; proposed milestones, not completed acceptance gates.

Make building, reviewing and accepting the work explicit

Role	Working responsibilities
Manish - Analyst	Sources evidence, develops calculations and records checks; available 100%.
Nigel - Modelling lead	Reviews methodology, priorities and interpretation.
Henry - Reviewer	Challenges the model before formal use.
Completion criterion	Required evidence
Traceability	A traceable source and assumption owner.
Calculation and check	An explainable calculation and a relevant check.
Review record	Documented limitations and review findings.

Team roles are confirmed. Agree review cadence and acceptance tolerances.

Keep delivery aligned through check-ins, reviews and client touchpoints

Proposed working rhythm | Monday HG check-in already planned; recurring slots and client dates to agree

Touchpoint	Frequency and participants	Required output
Internal check-in	Daily, 10–15 minutes; Manish + Nigel	Today's priorities, expected outputs, dependencies and blockers
Internal check-out	Daily written update; Manish to Nigel; call if blocked	Completed work and evidence links; checks, changes, open gaps and next steps
Technical review	Weekly and at model gates; Manish + Nigel; Henry at gates	Reconciled baseline, assumptions, calculation checks and integration evidence; review findings recorded
Partner review	Weekly proposed slot, before client checkpoints; partner + Nigel; Manish supports	Partner-led SCR and storyline; technical evidence, emerging implications, gaps and decisions needed
Client touchpoints	Proposed weeks 1, 3, 5 and 6; partner leads, model team as needed	W1: scope/data; W3: emerging findings; W5: draft implications; W6: final review and handover

Monday HG check-in: align the technical workstream, responsibilities and ways of working before delivery proceeds.

Agree the week-1 baseline, first corridor and acceptance criteria

Decisions for this session

1. Confirm baseline data and M/M assumptions: demand, domestic production and available imports.
2. Agree reporting checkpoints and reconciliation tolerances.

Week-1 baseline

Document historical demand, domestic production and imports. Produce a reproducible reference run and an initial reconciliation against Excel.

Baseline gate

Record inputs and assumptions; reproduce the run; explain material differences or assign owners and resolution dates. Nigel reviews the baseline; Henry confirms readiness to proceed.

Appendix

Evidence and model detail

2 October 2026

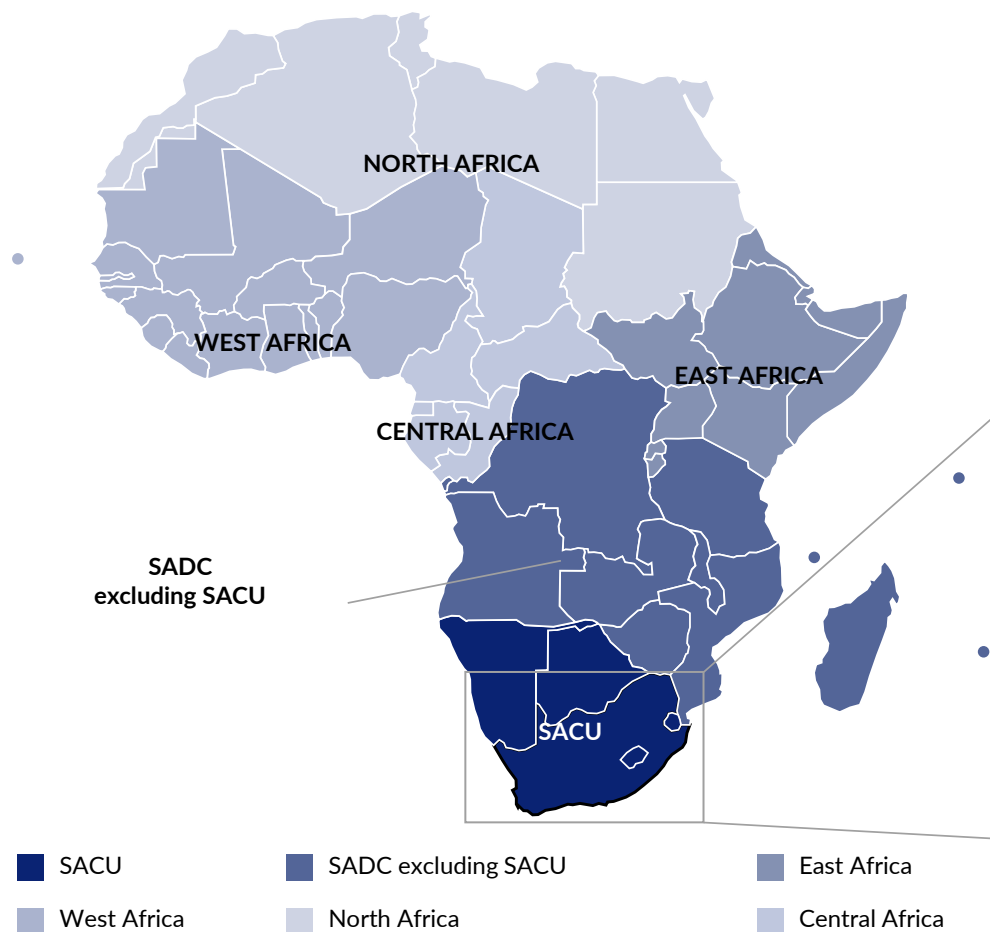
Internal working discussion

Photo source: Royal Vopak | Terminal Lesedi | vopak.com



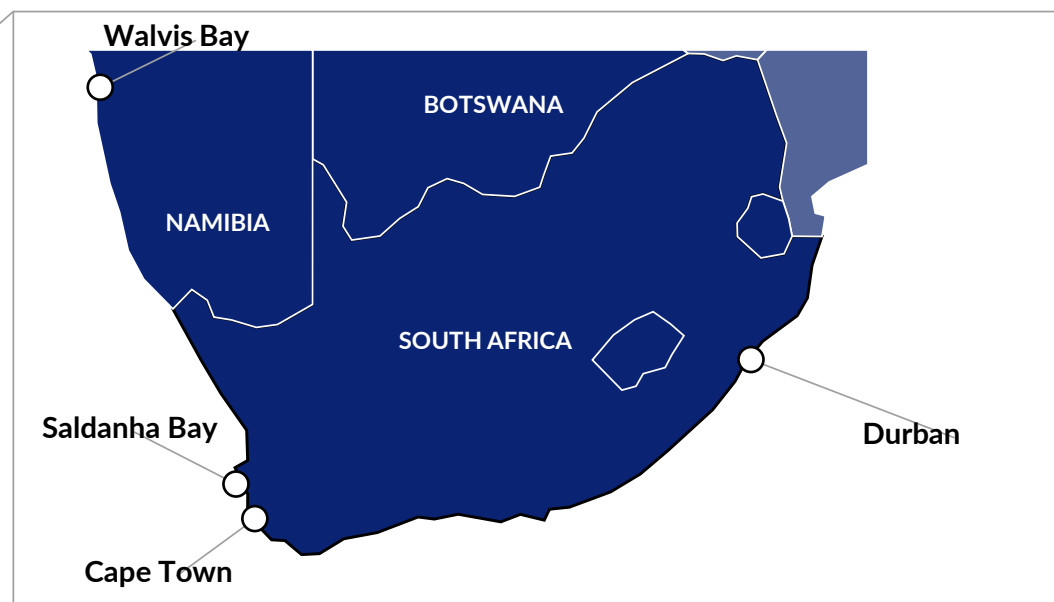
Separate current South African model coverage from wider African screening

Africa: agreed regional scope



South Africa: current model coverage

South African results appear on pp. 38–40 and 47–51. National demand and six production assets are covered; port allocations remain to build.



Colours = scope, not data readiness. SADC takes precedence over East/Central subregions; grey = other markets.

Locate the assets that produce South Africa’s liquid fuels

1 Production

2 Pipelines

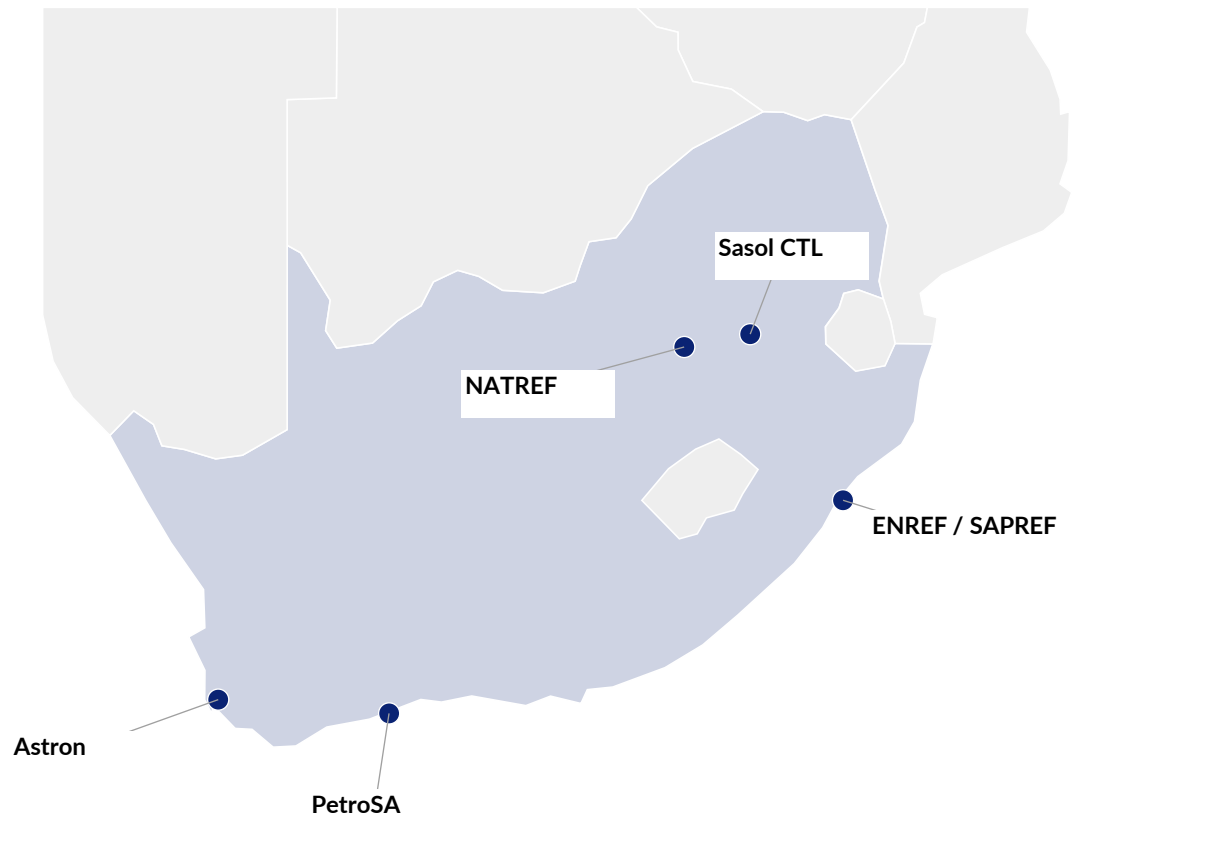
3 Ports

4 Roads

5 Rail

6 Storage

7 Corridors



Establish the domestic supply points

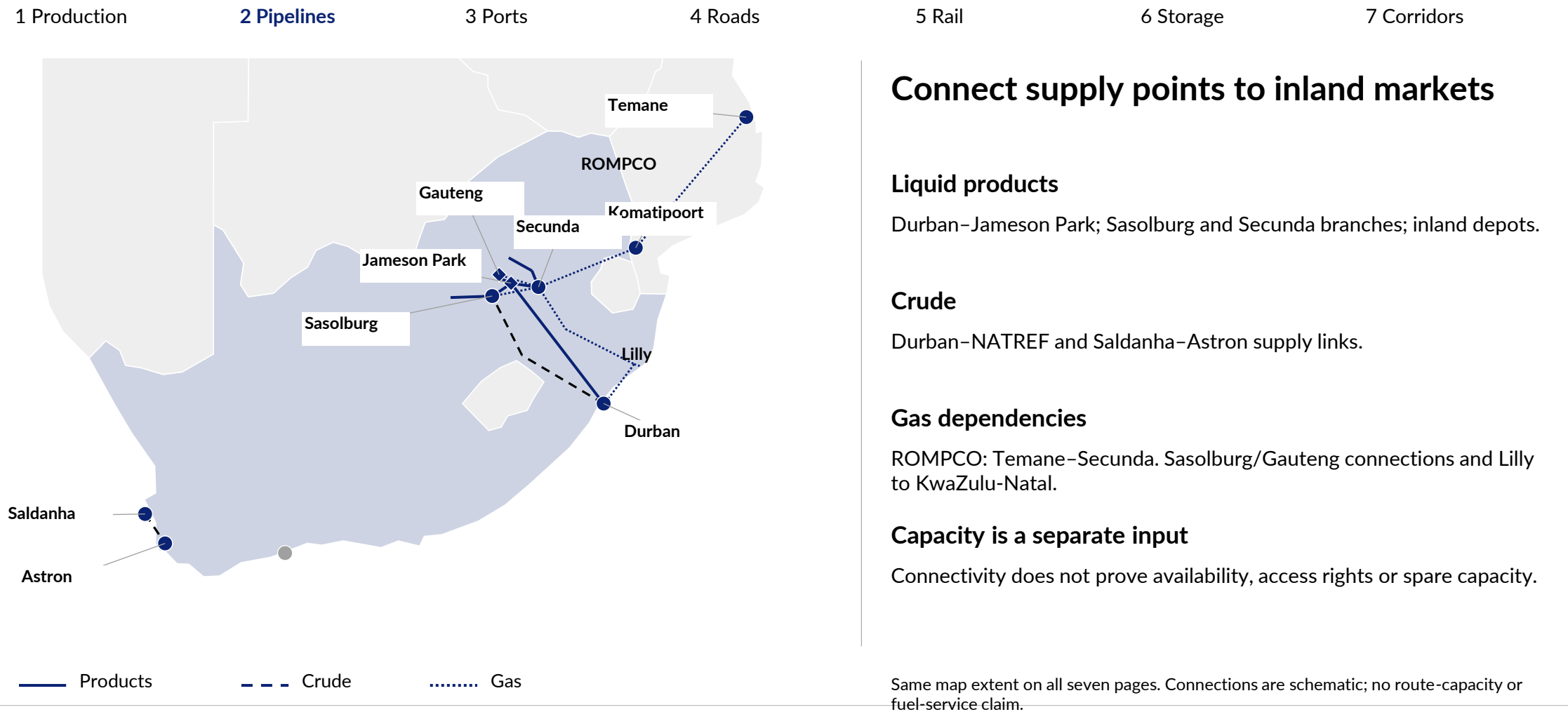
Asset	kbpd	Petrol	Diesel	Jet
ENREF	120	45%	25%	0%
SAPREF	180	45%	25%	10%
NATREF	108.5	45%	25%	10%
SASOL	75	85%	15%	0%
ASTRON	100	45%	25%	10%
PETROSA	45	85%	15%	0%

Workbook capacity and yield assumptions. Inclusion does not establish current operation; utilisation and availability determine model output.

Same map extent on all seven pages. Connections are schematic; no route-capacity or fuel-service claim.

● Production assets | approximate site locations

Trace pipeline connections between supply points and inland markets



Locate the ports through which imported fuel enters the system

1 Production

2 Pipelines

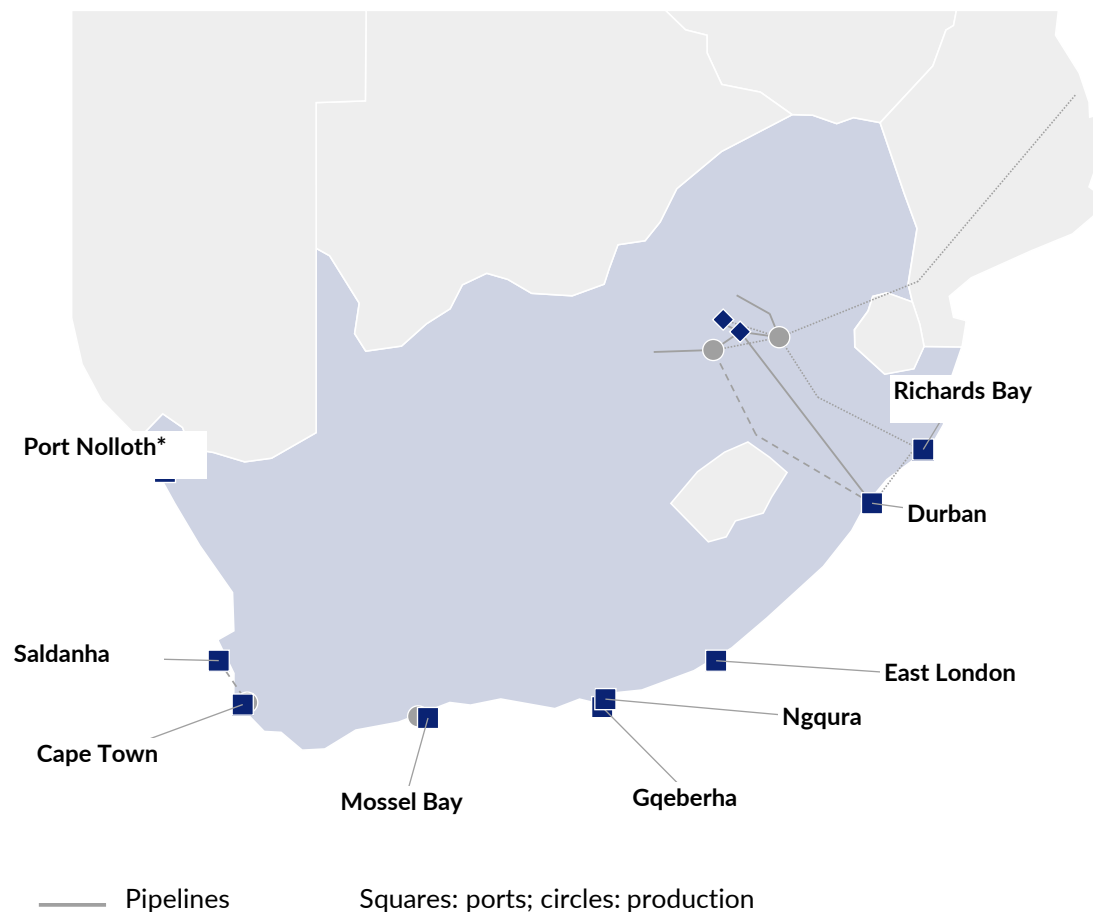
3 Ports

4 Roads

5 Rail

6 Storage

7 Corridors



Add the coastal import entry points

Eight commercial ports

Saldanha, Cape Town, Mossel Bay, Gqeberha, Ngqura, East London, Durban and Richards Bay.

Port Nolloth shown separately

Non-commercial port; inclusion does not imply fuel import service.

Test handling capability

Berths, draft, product compatibility, receipt rates, storage connections and access.

Allocate imports explicitly

National import requirements are not yet assigned to ports.

Same map extent on all seven pages. Connections are schematic; no route-capacity or fuel-service claim.

Add road connections to inland markets and border crossings

1 Production

2 Pipelines

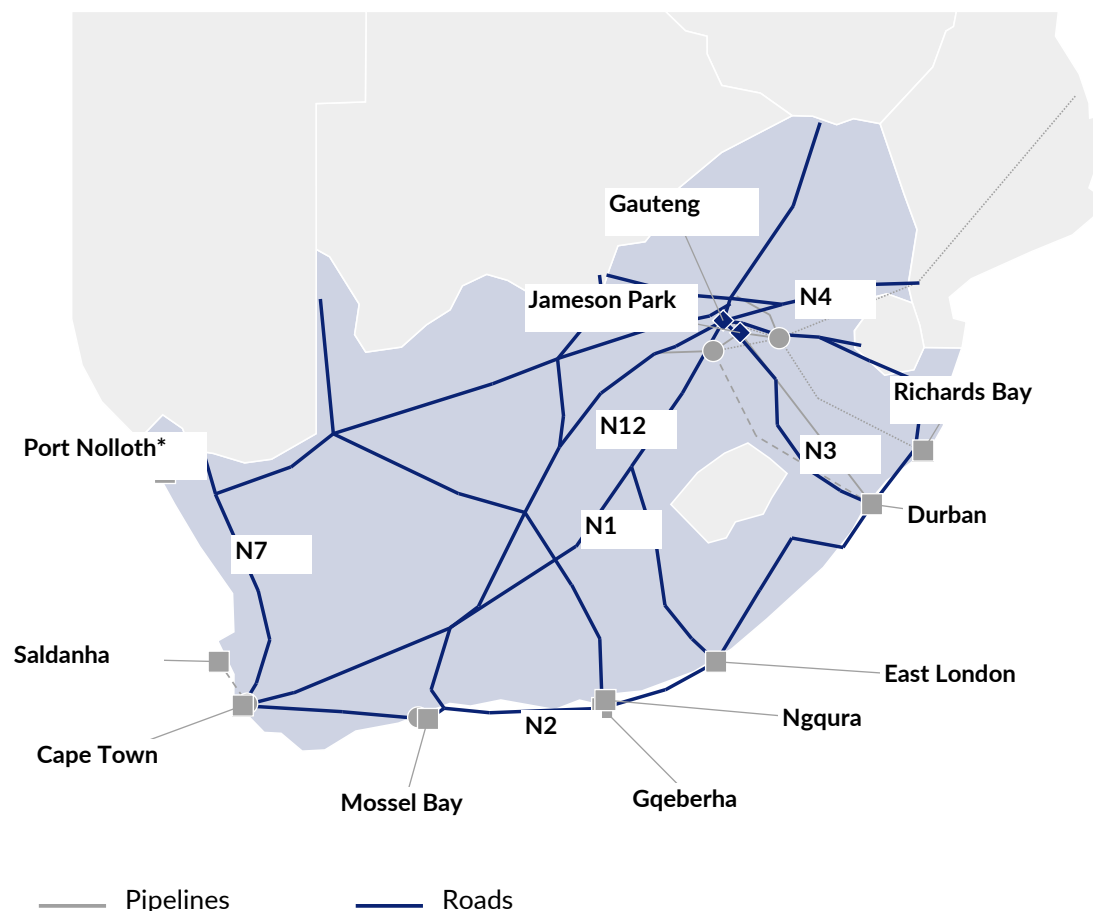
3 Ports

4 Roads

5 Rail

6 Storage

7 Corridors



Extend connections by road

Coastal to inland

N3 connects Durban with Gauteng; N1 links Cape Town and inland markets.

Regional and border links

N2, N4, N7 and other selected national corridors extend market reach.

Translate roads into fuel capacity

Tanker fleet, cycles, permissions, loading slots and receiving capacity are required.

Earlier layers remain context

Production, pipes and ports are subdued; selected roads are highlighted.

Same map extent on all seven pages. Connections are schematic; no route-capacity or fuel-service claim.

Add rail connections and identify where fuel service needs verification

1 Production

2 Pipelines

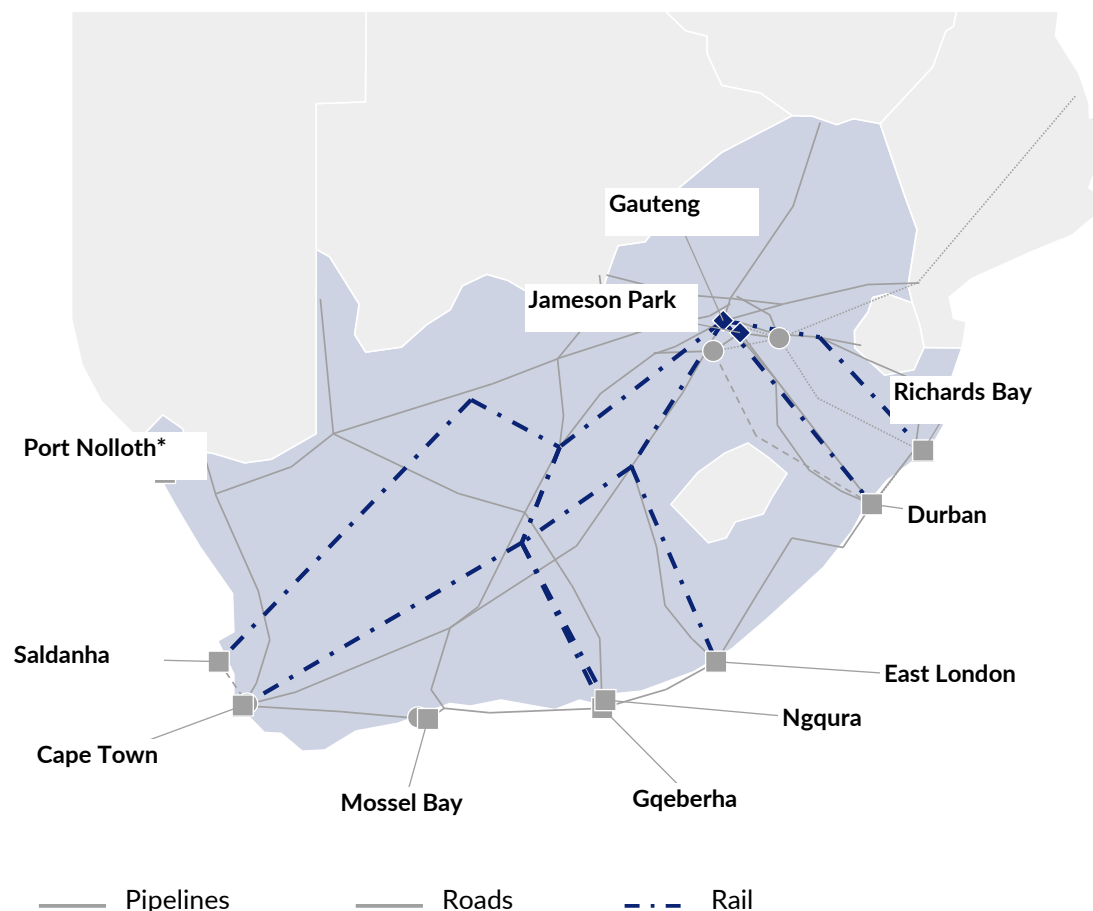
3 Ports

4 Roads

5 Rail

6 Storage

7 Corridors



Add rail and test actual fuel service

Coastal-Gauteng routes

Durban, Cape Town and Eastern Cape connections.

Other major network families

Sishen-Saldanha; Northern Cape-Nelson Mandela Bay; eastern hinterland-Richards Bay.

Network presence is not fuel service

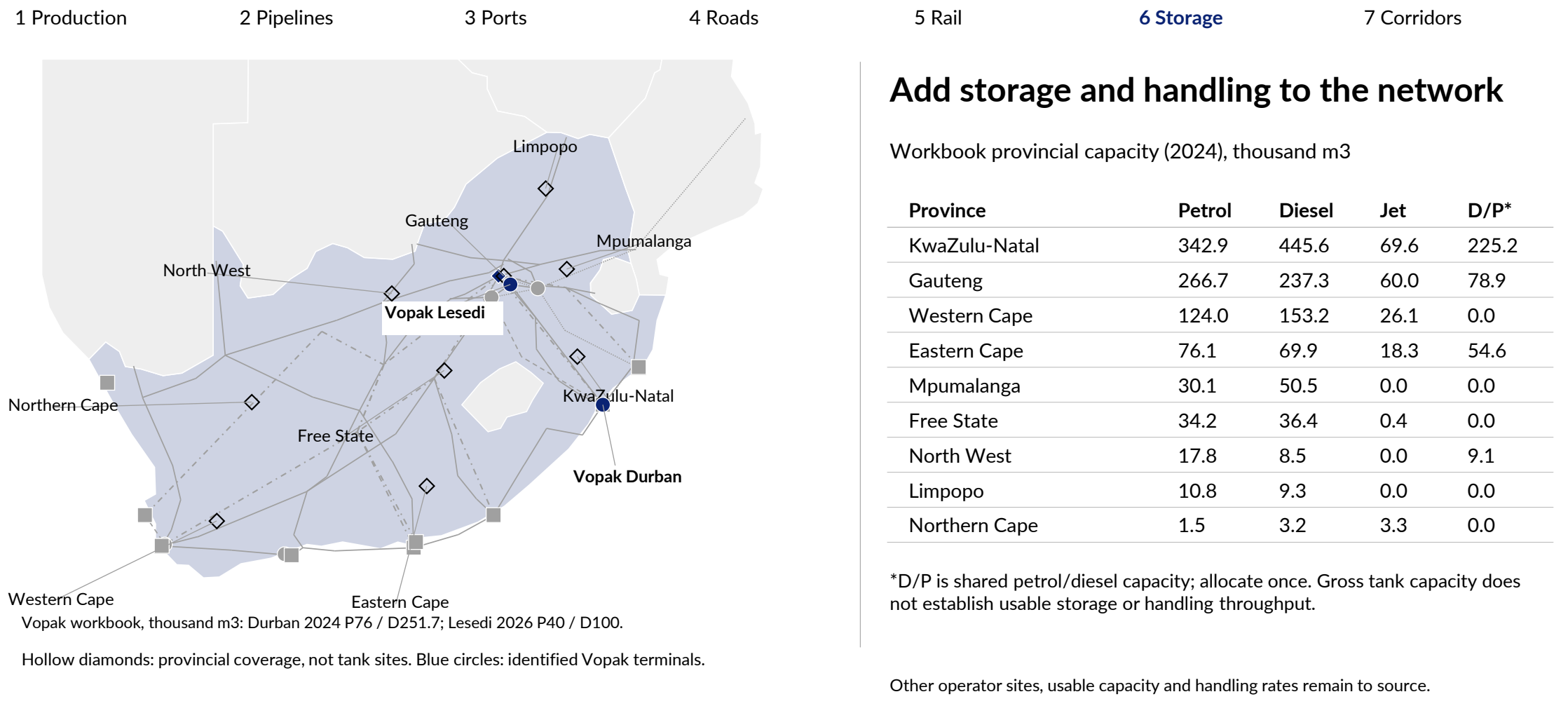
Confirm fuel wagons, slots, sidings, service reliability and terminal interfaces.

Avoid treating bulk routes as fuel routes

Iron ore and manganese corridors illustrate the network, not available fuel capacity.

Same map extent on all seven pages. Connections are schematic; no route-capacity or fuel-service claim.

Locate storage across operators and reconcile capacity by product



Bring the national network together into corridor families to assess

1 Production

2 Pipelines

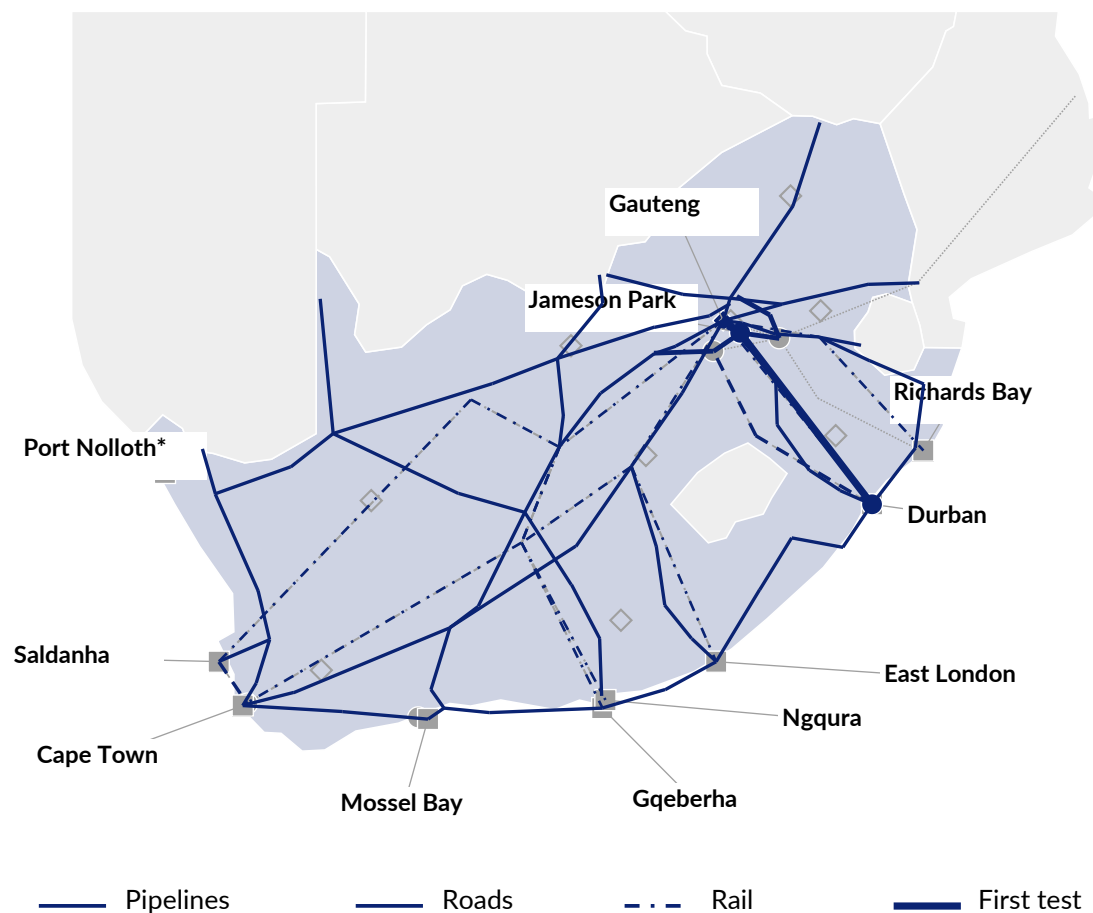
3 Ports

4 Roads

5 Rail

6 Storage

7 Corridors



National corridor families to assess

Durban - Gauteng

N3, NMPP and rail; first corridor to test.

Richards Bay - eastern hinterland

Ermelo / Mpumalanga and Gauteng connections.

Cape Town / Saldanha - inland

N1, N7 and rail; Saldanha - Astron crude link.

Mossel Bay - coast and inland

N2 / N12 connections; validate product service.

Gqeberha / Ngqura - inland

N10 and rail towards Northern Cape / Free State.

East London - Free State / Gauteng

N6 and rail connections; validate fuel service.

Northern and border approaches

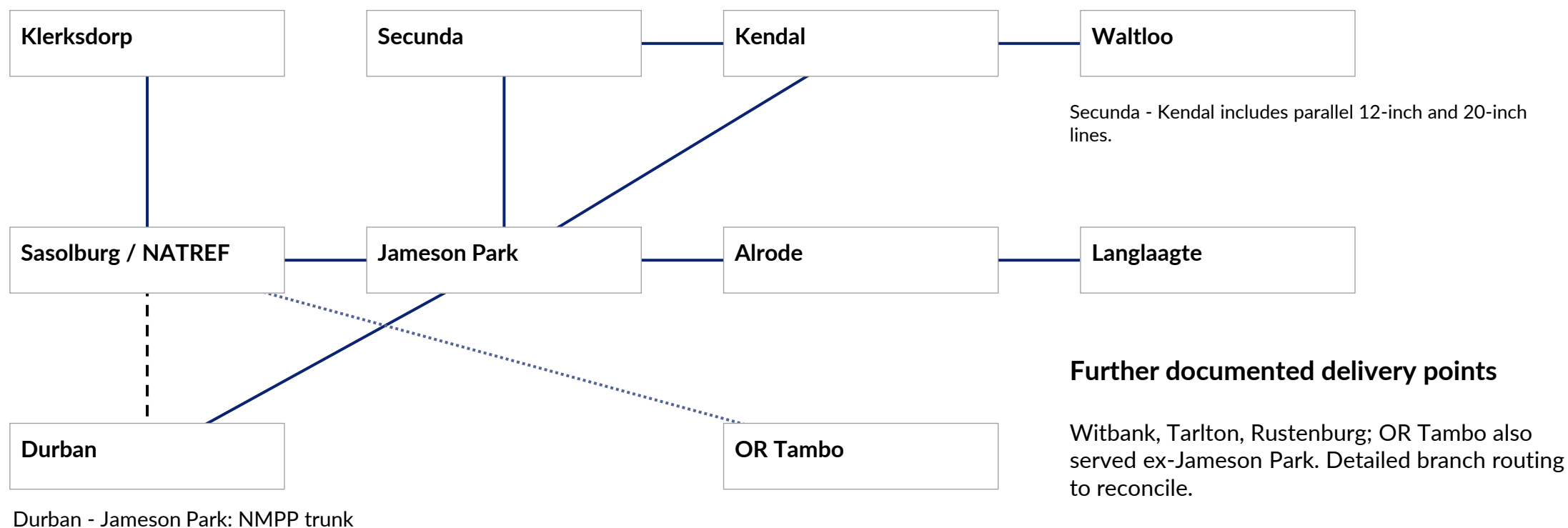
N1 north; N4 / N17 east-west; N7 / N14 / N18 west.

All eight commercial ports are represented. These are national screening families, not validated fuel flows or an exhaustive local-route inventory.

Same map extent on all seven pages. Connections are schematic; no route-capacity or fuel-service claim.

Trace the documented liquid-pipeline links and delivery points

Liquid-pipeline connections | network schematic, not geographic routing



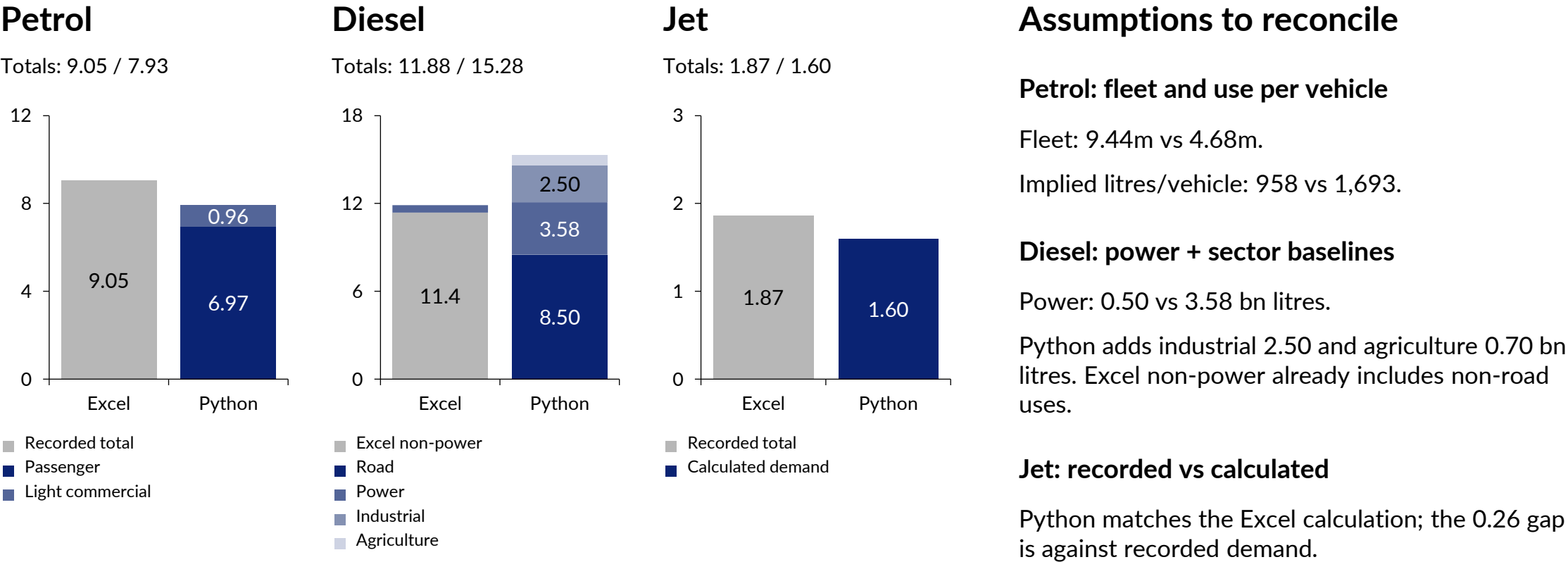
Western Cape: Saldanha - Astron refinery crude pipeline

— Products - - - - Crude Jet

Gas: see ROMPCO / Sasol / Lilly page.

Reconcile the 2024 baseline by fuel and demand component

South Africa, 2024 | billion litres | current high_demand configuration



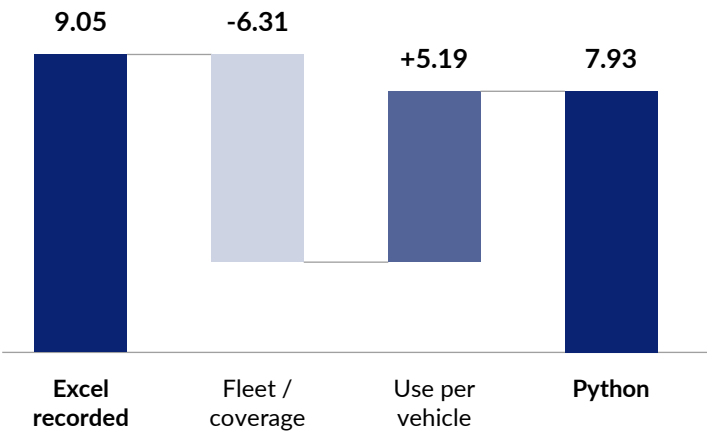
Excel aggregates remain unsplit where sector mapping is unresolved. Panel scales differ; driver bridges follow.

Trace the components behind the Excel–Python demand differences

2024 | billion litres | Excel recorded total → reconciliation components → Python total

Petrol

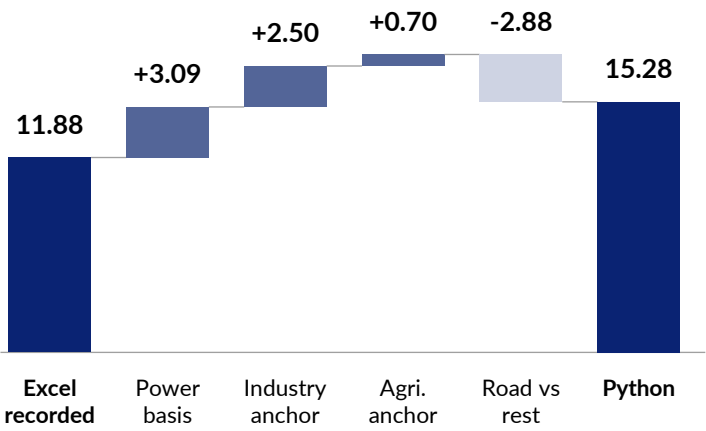
Net difference: -1.12 bn L



Fewer vehicles: -6.31 bn L.
Higher use per vehicle: +5.19 bn L.
Check fleet coverage, mileage and economy.

Diesel

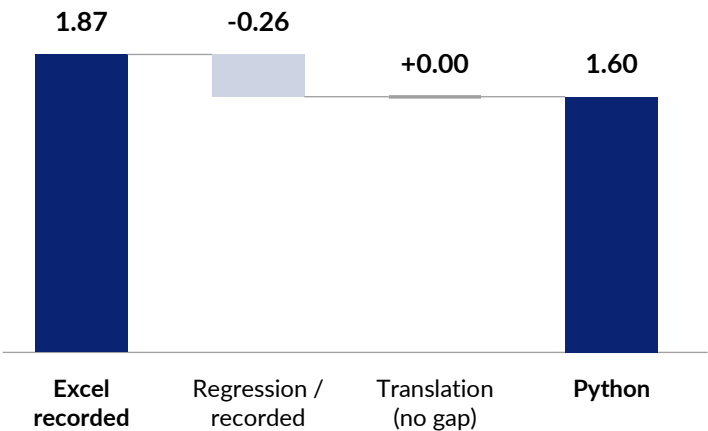
Net difference: +3.41 bn L



Power and sector anchors increase demand.
Road/non-power definitions partly offset this.
Validate dispatch and possible overlap.

Jet

Net difference: -0.26 bn L



Python matches Excel’s calculated demand.
The gap is against recorded demand.
Review regression fit and fuel definitions.

Bar labels show totals or signed changes; + adds demand, – reduces it. Each fuel has its own scale. Bridges reconcile totals; they do not prove causation.

Validate the evidence before choosing a reference model

Use Excel as a provisional historical reference; validate Python before relying on its forecasts.

Test	Excel evidence	Python evidence	Assessment / next check
Petrol	9.05 bn litres recorded; 9.44m vehicles. Litres/vehicle is inferred from the total. [1]	7.93 bn litres; cohort fleet 4.68m. Mileage, fuel use and scrappage include provisional inputs. [2]	No accuracy winner. Reconcile fleet coverage and validate kilometres and fuel consumption.
Diesel	11.88 bn litres total; power 0.50; remaining 11.38 includes non-road demand. [1]	15.28 bn litres; power 3.58 plus industrial 2.50 and agriculture 0.70 placeholders. [2,3]	Python uplift is not validated. Check power history and sector overlap before accepting it.
Jet	Recorded demand 1.87; regression calculates 1.60 bn litres. [1]	Same regression result: 1.60 bn litres, matching Excel to numerical precision. [3]	Translation agrees; accuracy is unproven. Check product coverage and fit to observed demand.

Decision gate: compare both against independent, like-for-like actuals across several historical years; then test forecasts on held-out years. [4,5]

[1] Inherited workbook: Gasoline - DemandSupply K29/T29/X29; Diesel - DemandSupply K30/T30/M30; Jet - DemandSupply J23; fJetFuel H24.
[2] assumptions/2026/vehicles.yaml, generation.yaml, industrial.yaml, agriculture.yaml; src/lfm/model/demand/vehicles.py.
[3] src/lfm/reporting/reconciliation.py; pptx/output/data/driver_reconciliation.json (2024, high_demand).
[4] GATE_CHECKLIST.md: independent review and business approval outstanding. [5] DMRE SA Fuel Sales Volume portal: external benchmark to reconcile, not yet validated here.

Validate the existing macro and sector inputs before extending coverage

Data family	Already in the repository	Validation or extension needed
Macro drivers	GDP, GDP per capita and population: 2005-2050; GDP growth: 2020-2050	Separate history from projections; refresh sources; FX remains blank
Demand benchmarks	Petrol, diesel and jet history in one CSV; comparison uses years through 2024	Reconcile totals, fuel definitions and provisional sector baselines
Transport and power	Vehicle assumptions; efficiency series 2020-2050; passengers 2017-2050; OCGT series	Review opening fleet, adoption, efficiencies and power-sector assumptions
Domestic production	Six South African refinery / synfuel assets; series 2008-2050; capacity and product mix	Check operating status, utilisation and forecast assumptions
Geographic extension	Structured time series currently cover South Africa	Source other SACU countries and priority African screening data

Available inputs are not yet validated inputs. Date ranges combine historical and forecast entries.

Centralise and validate the infrastructure and market-access evidence

Evidence family	Material already held	Dataset to assemble and validate
Ports and terminals	Excel: Durban and Lesedi capacity / throughput; report: Walvis Bay and competitor evidence	Reconcile site, operator, fuel, capacity, throughput and vintage; add missing ports
Inland corridors	Prior report: Trans-Kalahari, Trans-Cunene and Walvis Bay-Ndola-Lubumbashi routes	Origin-destination, mode, capacity, cost, transit time and constraints
Imports and market access	Prior report: import trends, competitors and potential regional offtakers	Trade by product and route; feasible supply, access rights and contracts
Storage requirements	Excel: provincial fuel storage, terminal capacities and tank turns	Validate usable capacity, stock cover, utilisation, source dates and commitments
Regional opportunities	Namibia / Walvis Bay assessment; proposal calls for wider African screening	Comparable screening data for East, West, Central and North Africa and SADC excluding SACU

Centralise WS5 evidence in the master assumptions registry; consume a pinned, registered snapshot. Local Excel entries remain unvalidated and unwired.

Match analytical depth and client outputs to the agreed market scope

Core calculation chain	Current engine: South African demand and domestic production; paired H/L runs. Target: available imports, nine independent H/M/L cases and M/M baseline. Remaining gaps require deliverable-import evidence.
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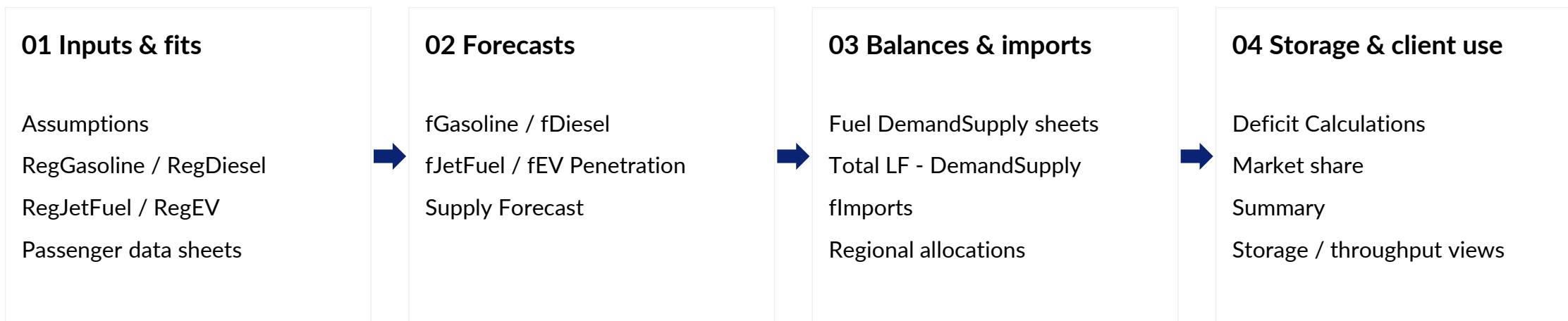
Market coverage and data	SACU, with a Namibia/Walvis Bay focus; screen East, West, Central and North Africa and SADC excluding SACU. Gather consumption, production and trade data, plus port, storage and corridor evidence.
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Vopak reporting and analysis	Board outlook: balances, imports, sensitivities and changes since the previous study. Assess logistics, market access, competing terminals, Walvis Bay storage and priority African opportunities.
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Keep Vopak-specific commercial assumptions and interpretations separate from reusable calculations.

Trace how Excel turns assumptions into forecasts, balances and storage views

Inherited workbook | sheet-level calculation and reporting structure



Scenario logic and limitations

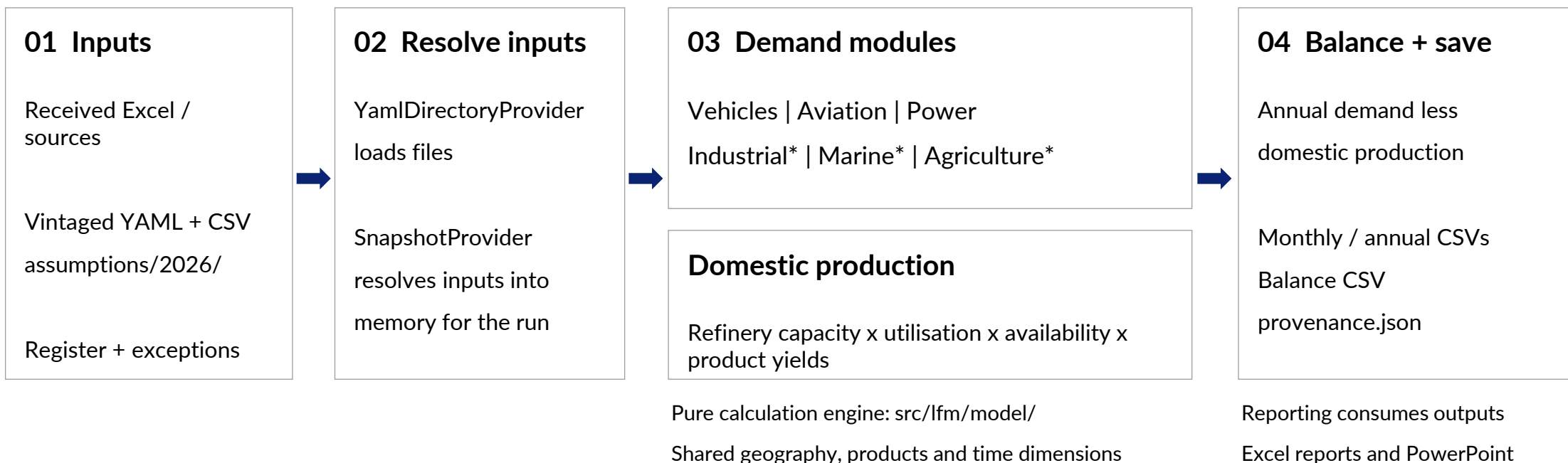
High/low demand and production assumptions feed workbook formulas; this is not yet the independent nine-case H/M/L matrix.

fImports contains external workbook references. Cached values, regional allocation formulas and storage/market-share assumptions need reconciliation.

Trace how Python resolves inputs, calculates balances and records each run

One run = code version + assumption vintage + scenario

CLI coordinates the run, checks declared inputs and records provenance | `src/lfm/cli.py` + `run.py`

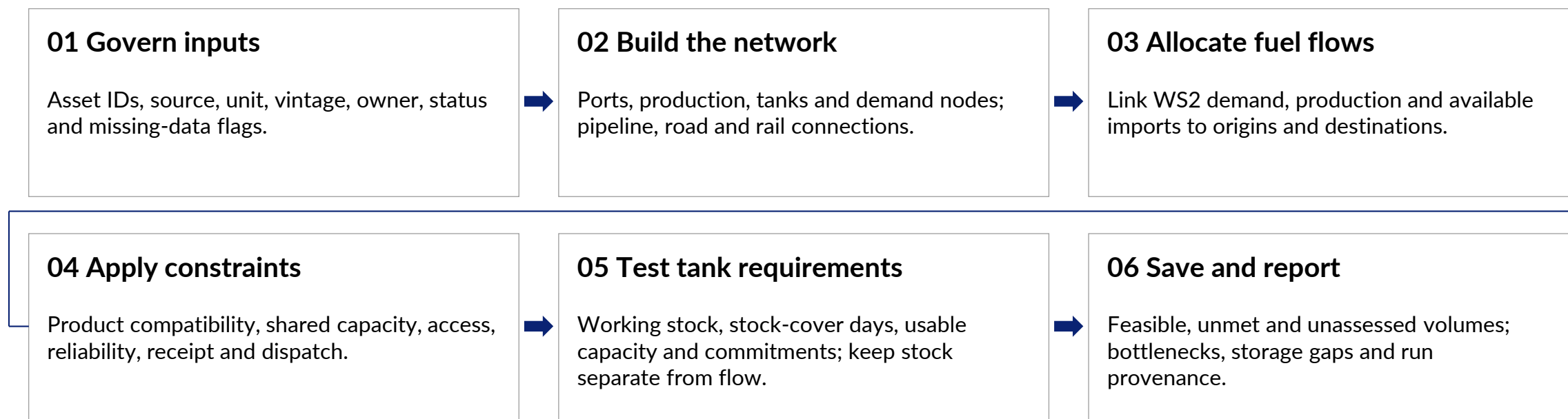


*Held sectors use provisional baselines scaled by GDP. Current data: South Africa; scenarios: high_demand / low_demand.

Target extensions: independent H/M/L scenarios, available imports and WS5 infrastructure; see the Model Build section.

Trace how infrastructure inputs become feasible flows and storage requirements

Target architecture | inputs and source material exist; allocation and constraint engine still to build



Checks across the chain: node conservation • shared-capacity accounting • manual corridor tie-out • scenario response

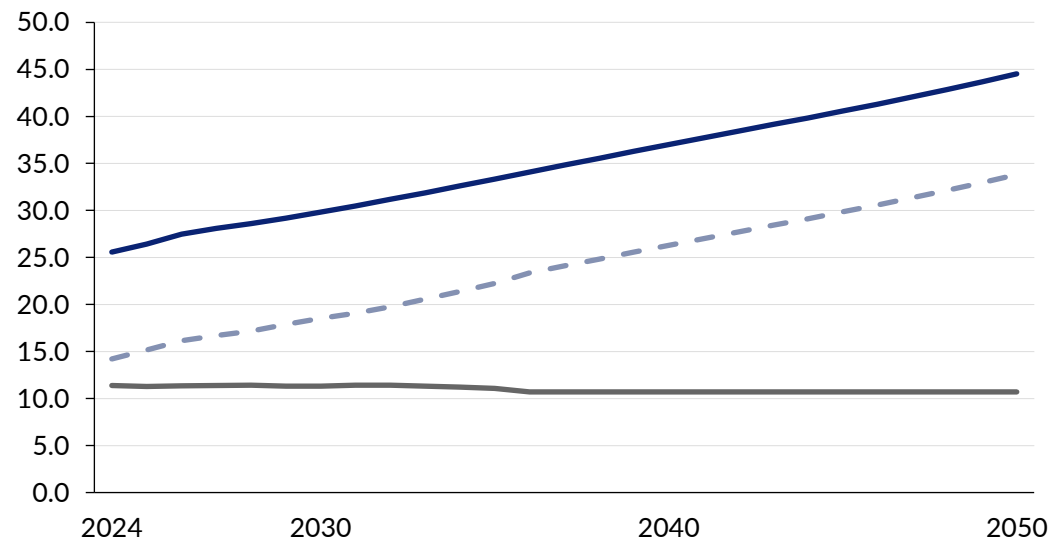
Power and gas dependencies inform availability; airports and power stations receive fuel-demand allocations. Commercial interpretation remains in WS3.

Compare demand with domestic production to quantify the gap before imports

South Africa | 2024–2050 | billion litres per year | current paired scenarios; provisional

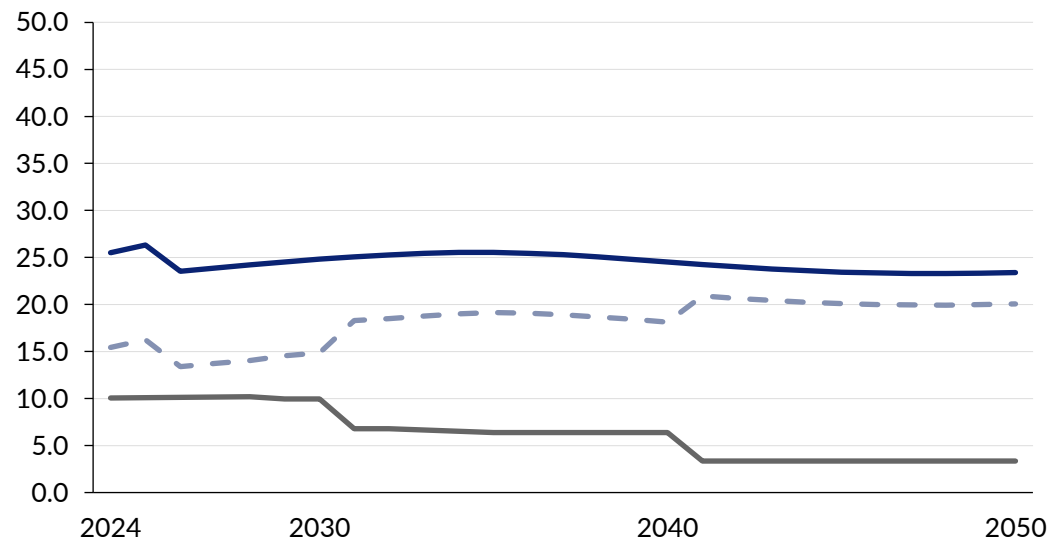
— Demand — Domestic production - - - Deficit

High Demand



2050 deficit: 33.8 bn litres

Low Demand



2050 deficit: 20.1 bn litres

Core total: petrol + both diesel grades + jet; excludes fuel oil.

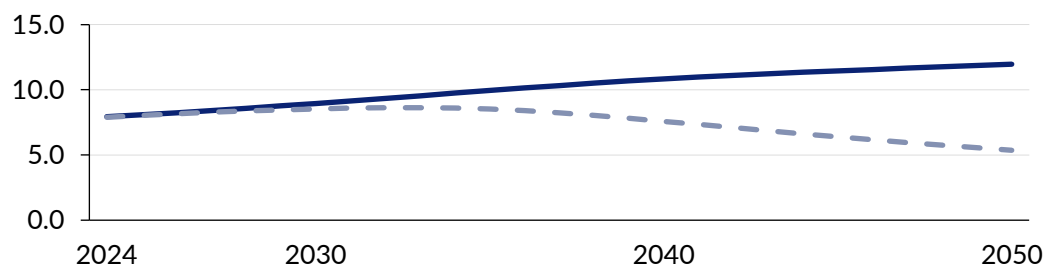
Deficit = demand – domestic production; not unmet demand after imports. Trade, stocks and import constraints remain unresolved.

Compare petrol, diesel and jet demand across the current scenarios

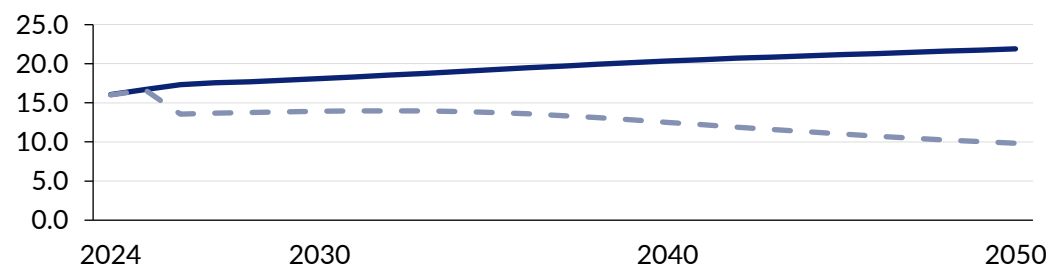
South Africa | 2024–2050 | billion litres per year | current paired scenarios; provisional

High-demand run Low-demand run

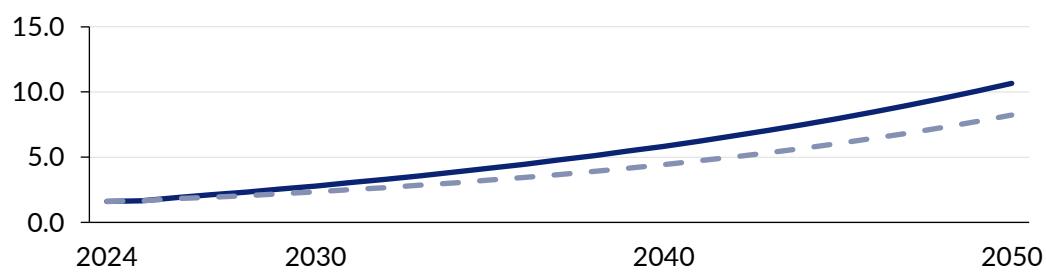
Petrol



Diesel – both grades



Jet



Reporting scope

Petrol, diesel (both grades) and jet.

Fuel oil is a separate provisional marine output; see the supplementary page.

Axes differ by fuel. Core totals exclude fuel oil; no numerical model assumptions have changed.

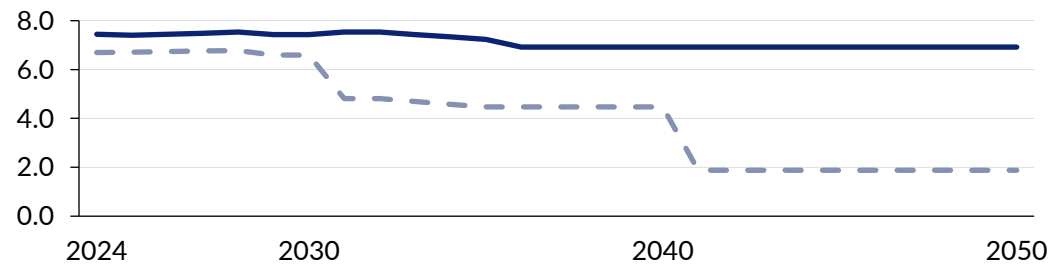
Deficit = demand – domestic production; not unmet demand after imports. Trade, stocks and import constraints remain unresolved.

Compare domestic production by fuel across the current scenarios

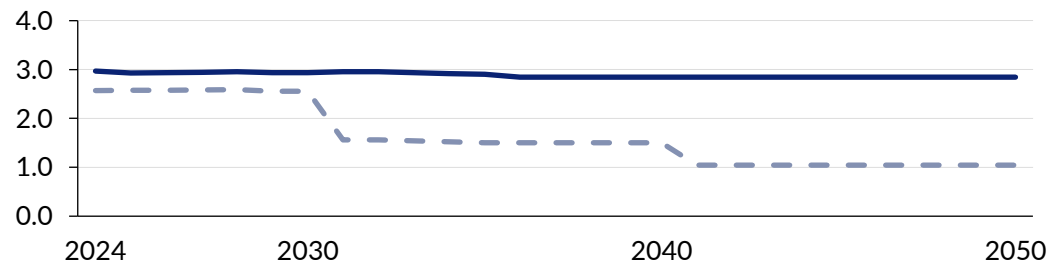
South Africa | 2024–2050 | billion litres per year | current paired scenarios; provisional

High-demand run Low-demand run

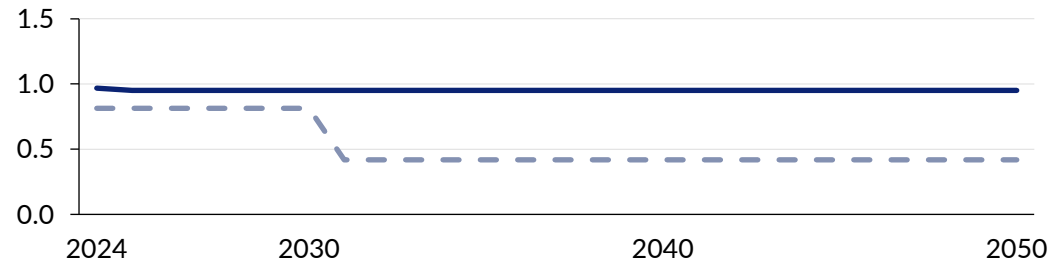
Petrol



Diesel – both grades



Jet



Reporting scope

Petrol, diesel (both grades) and jet.

Fuel oil is a separate provisional marine output; see the supplementary page.

Axes differ by fuel. Core totals exclude fuel oil; no numerical model assumptions have changed.

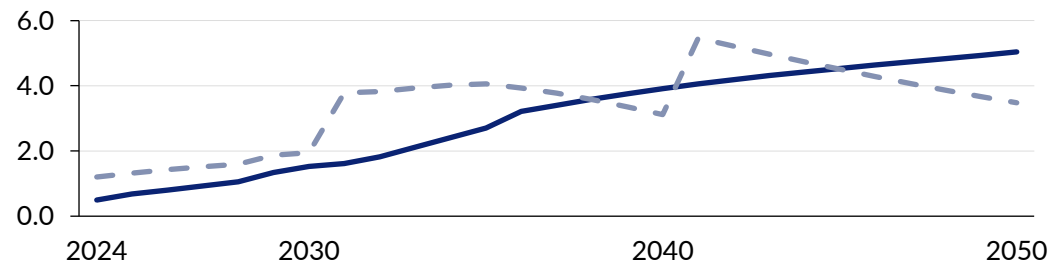
Deficit = demand – domestic production; not unmet demand after imports. Trade, stocks and import constraints remain unresolved.

Identify when and where domestic production falls short of demand

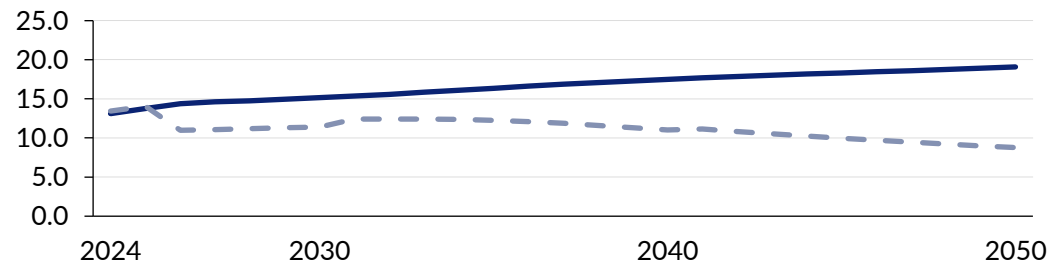
South Africa | 2024–2050 | billion litres per year | current paired scenarios; provisional

High-demand run Low-demand run

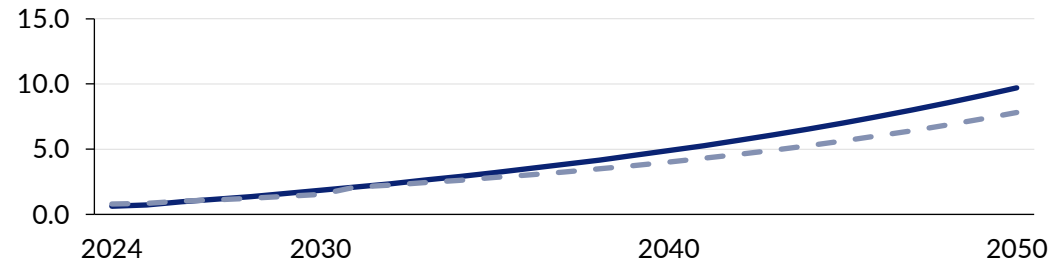
Petrol



Diesel – both grades



Jet



Reporting scope

Petrol, diesel (both grades) and jet.

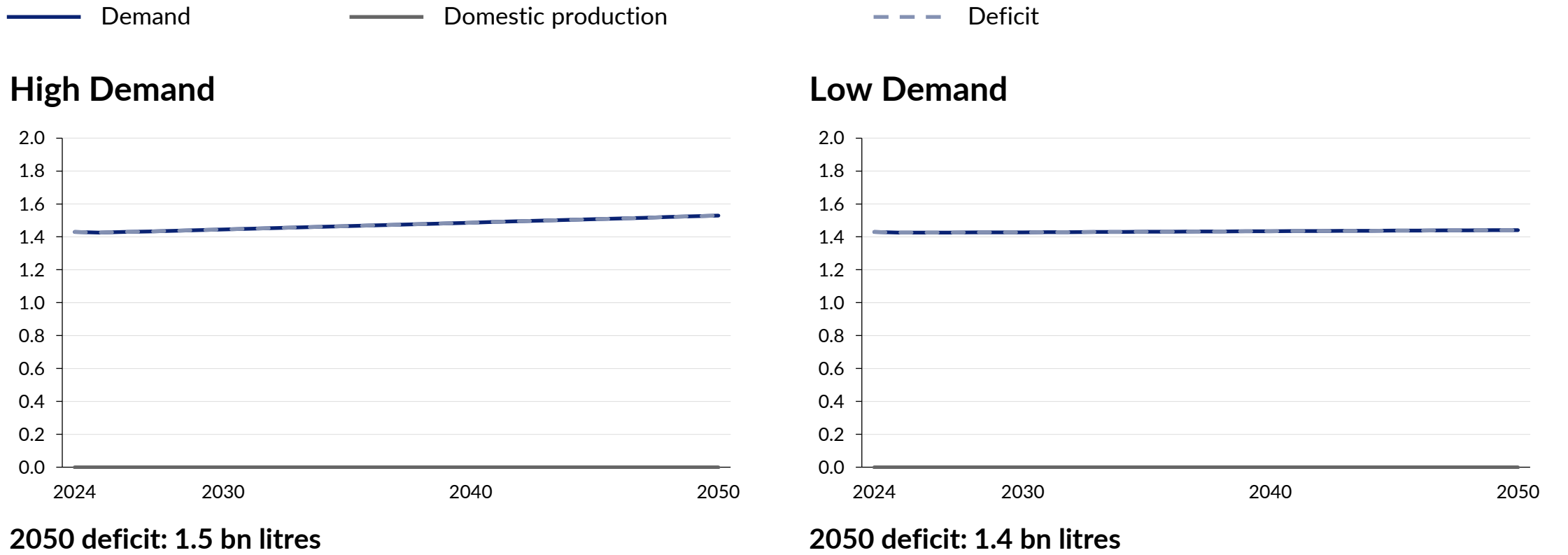
Fuel oil is a separate provisional marine output; see the supplementary page.

Axes differ by fuel. Core totals exclude fuel oil; no numerical model assumptions have changed.

Deficit = demand – domestic production; not unmet demand after imports. Trade, stocks and import constraints remain unresolved.

Review marine fuel oil separately from the core petrol, diesel and jet scope

South Africa | 2024–2050 | billion litres per year | current paired scenarios; provisional



Supplementary fuel oil only: provisional marine baseline; no domestic production represented.

Deficit = demand – domestic production; not unmet demand after imports. Trade, stocks and import constraints remain unresolved.

Agree priorities and next steps

Discussion

2 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Lesedi | vopak.com

